

A Vision Transformer based Deep Learning Approach for Brain Tumour Classification from MRI: An Explainable AI Framework

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Abstract—Brain tumors are a life-threatening disease and their invasiveness requires early and accurate diagnosis which in turn will increase the survival of the patient. Although Magnetic Resonance Imaging (MRI) is an essential diagnostic tool, manual interpretation can be very time consuming and subject to human error because there are always small variations in the appearance of the tumor. The proposed model overcomes these problems by suggesting a robust automated classification design that uses the Vision Transformer (ViT) architecture. The dataset used in the methodology consisting 7,023 MRI images classified as gliomas, meningioma, pituitary, and no tumor. The proposed ViT model is designed to learn global spatial interactions with the help of self-attention. A 5-fold cross-validation approach has been carried out so as to stabilize the statistical features at the end of the training pipeline. The accuracy of the ViT approach has proven to be much higher, it reached the maximum training accuracy of 99.94 and the testing accuracy of 98.51, performing better than the several traditional models, like ResNet-50, Alex-Net, and DenseNet-201. Further to address the black-box nature of deep learning models, a Grad-CAM and LIME based explainable artificial intelligence (XAI) are integrated. This approach enables the visual interpretation of the model's decision-making process. The Grad-CAM and LIME results highlight that the model focuses on clinically relevant regions and ignore irrelevant background features. This enhances the transparency, reliability, and clinical applicability of the proposed method.

Keywords—Brain tumor, deep learning, image processing, magnetic resonance imaging, vision transformer.

I. INTRODUCTION

Brain tumour is known as one of the most serious and life-threatening neurological problems. They are normally caused by the abnormal and uncontrolled proliferation of brain cells that normally form part of the body, disabling the key body functions as well as causing neurological impairments such as movement, sensation, and thinking. Medical science has traditionally classified these tumours into three major groups namely gliomas, meningioma and pituitary tumours [1]. The burden of these conditions is substantial worldwide, in 2024, approximately 308,102 new cases and 251,329 fatalities were due to brain and central nervous system (CNS) cancers [2].

Each type has its own growth patterns and treatment difficulties. Gliomas are especially high-grade glioblastomas, are the most aggressive and spread rapidly through the brain tissues of a person, often evading our body's immune defences which make them difficult to treat and reduces the survival rates. As the number of brain cancer cases increases continuously worldwide, it has become one of the leading causes of cancer-related deaths worldwide which highlights the need for accurate and early diagnostic methods. MRI plays crucial role in detecting brain abnormalities and gives high resolution, non-invasive images of brain. However, analysis of these images manually takes a lot of time which can be affected by fatigue and is limited by minor or small differences in tumour appearance which increases the risk of misdiagnosis or delayed treatment [3]. Over recent years, there has been a call in the clinical community to invest in diagnostic systems that are able to not only accelerate the identification of the tumour but across various imaging settings and patient populations remain consistent. Although useful, traditional radiological examination struggles to keep up with the booming growth of medical imaging and the consequent complexity of tumour shapes. This has demonstrated the necessity of computational tools which can be used to process large volumes of data, identify subtle anomalies that humans may not, and deliver objective and repeatable evaluations. The automated systems demonstrate how deep learning-based techniques can reduce the role of human intervention and enhance image quality [4]. The introduction of smart systems into the clinical workflow is not merely a technological addition to the work process, it is a practical necessity of contemporary healthcare. The objective of this change is to ensure that the gaps in expertise are eliminated, the disparities in the interpretation across observers is minimized and the radiologists can make quicker and more confident decisions when it comes to diagnostic judgment [5]. Fig. 1 illustrates the overall flowchart of the proposed methodology.

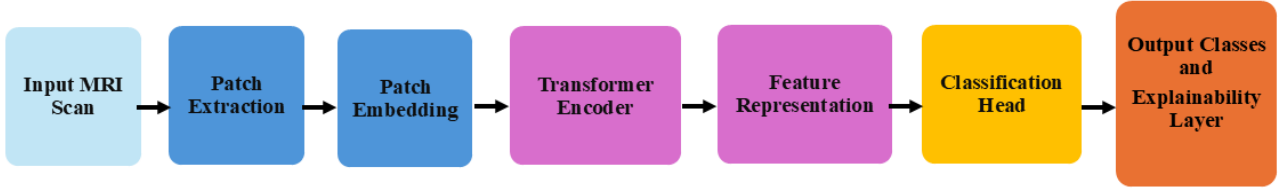


Fig. 1. Flowchart of the proposed VIT architecture

II. LITERATURE SURVEY

In recent years, machine learning (ML) and deep learning (DL) have started to change how brain tumors are detected. Models such as convolution neural network (CNN) based hybrids, attention-based models and transformer models now are capable of operating directly on MRI scans and can learn patterns on their own. Sahoo *et al.* [6] proposed a hybrid approach that integrates pruning techniques with EfficientNetV2B3 and metaheuristic optimization methods for feature selection. Binish *et al.* [7] introduced CNN framework integrating attention mechanisms and multi-resolution kernels. Soomro *et al.* [8] presented an approach based on machine learning and deep learning methods for brain tumor segmentation. Rinesh *et al.* [9] developed a hybrid framework using hyperspectral MRI along with clustering and optimization techniques for tumor detection. Ahmed *et al.* [10] implemented a multi-modal approach to AI, which comprises self-supervised learning and the data augmentation by the use of generative adversarial networks (GANs). The problem of the deficiency of the data was overcome by the methodology and contributed to the improvement of the learning of features and the consistency of the diagnosis. The consecutive fusion attention transformer was suggested by Goceri *et al.* [11] have the potential to be utilized to precisely classify the brain tumor. It was found that the model significantly employed local and global feature learning to achieve acceptable segmentation achievement. Hemanth *et al.* [12] suggested a deep CNN framework which minimizes the computational cost since it eliminates training by iterative fully connected layers. In the study by Razzak *et al.* [13], a cascaded two path way group CNN was introduced, accompanied by the addition of group convolutions to keep in mind the symmetry and multi-scales. The model improved the precision of segmentation and false positives in the tumor localization.

Light CNN were being proposed by Abiwinanda *et al.* [14]. The model demonstrated that a competitive performance could be achieved with simple architectures with reduced parameters and which are functionally efficient enough to be applied in real-life clinical applications. The brain tumor classification studied by Yang *et al.* [15], where transfer learning-based model like Alex-Net and Google-Net were being implemented. The cheaper Google-Net model has worked better and also demonstrated an encouraging outcome in multiple feature extraction and enhanced generalization. Vanilla CNN proposed in the article by Usman *et al.* [16] to classify brain tumors. The model demonstrated that simple architectures, which are enhanced in terms of input resolution, can obtain tumor features and give a reliable performance. Alhassan *et al.* [17] proposed a hybrid deep learning model for brain tumor classification employing bidirectional long short-term memory (Bi-LSTM)

network. Ullah *et al.* [18] have suggested a Bayesian based DeepLabV3 network that comprises uncertainty modelling and multi-scale feature combination. Hmidi *et al.* [19] came up with an optimized CNN with batch normalization and dropout to classify brain tumors into various classes. Sadad *et al.* [20] have designed a framework to classify tumors in multi-class with the implementation of the concept of the transfer learning strategy involving the utilization of deep learning on the ResNet50.

Based on the above literature, it can be observed that most existing approaches for MRI image analysis rely on traditional ML and DL models, particularly CNN-based architectures and hybrid frameworks. Although these methods achieve promising performance, they predominantly operate as black-box models, offering limited interpretability of their decision-making process, which is a critical requirement in clinical applications. In addition, CNN-based approaches mainly focus on local feature extraction and may not effectively capture global contextual relationships within MRI images. To address these limitations, this study proposes VIT based framework that leverages self-attention mechanisms for enhanced global feature representation. Furthermore, to overcome the black-box nature of existing models, the proposed approach integrates XAI namely Grad-CAM and LIME, to provide transparent and interpretable predictions. This combination aims to improve both model performance and trustworthiness in MRI based diagnostic systems.

III. MATERIALS AND METHODS

A. Database and Preprocessing

The brain tumor MRI dataset is used in this study, are collected from [21]. The dataset consists of 7023 MRI images of various patients, divided into training and testing sets to enable guided learning. These are categorized to four classes i.e. gliomas tumor, meningioma tumor, pituitary tumor, and no tumor. The four class tumor images are shown in Fig. 2.

To meet the model's pre-requisites, all of the images are resized to 224×224 pixels and then are normalized using ImageNet's mean and standard deviation values. To make the model more general and less likely to overfit, the training data is changed by adding random horizontal flipping, rotation ($\pm 10^\circ$), and brightness or contrast changes. The class wise image count of the dataset remains same after the augmentation and does not change after the augmentation of data that is pituitary: 1269, gliomas: 1240, no tumor: 1289 and meningioma: 1242. Then, 5-fold cross validation is used

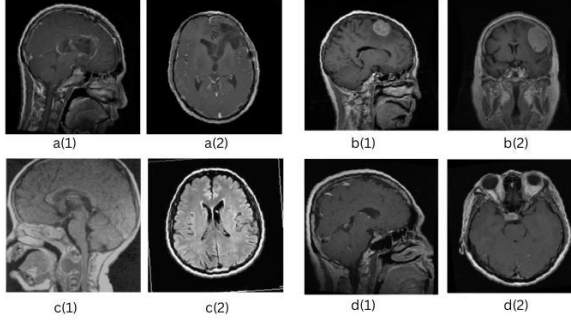


Fig. 2. MRI samples: (a) Glioma tumor (b) Meningioma tumor (c) No tumor (d) Pituitary tumor.

to make sure that the performance evaluation is accurate. To make training more efficient, data is loaded in mini-batches of size 64.

B. Overview of Vision Transformer (ViT)

The ViT is a deep learning architecture based on transformers, created for image classification tasks. Unlike CNNs, which uses convolution to extract features, ViT converts an image into a series of patches. It helps in computing the relationships between these patches with self-attention mechanisms. This approach helps the model to understand long-range connections across the complete image. This is especially useful for complex medical images like brain MRI scans. The complete data flow, from the input MRI image to the final prediction, is illustrated in Fig. 3.

In ViT architecture, an input image $I \in \mathbb{R}^{H \times W \times C}$ is first divided into fixed-size patches of dimension $P \times P$. The total number of patches N obtained from the image is calculated as:

$$N = HW \frac{HW}{p^2} \quad (1)$$

Each patch is flattened into a vector and projected into a lower-dimensional embedding space using a linear projection layer. Next, these patch embeddings are combined with positional embeddings to maintain spatial information about the original image structure. The resulting sequence of embeddings is then sent into a stack of transformer encoder layers [22].

To facilitate classification, a learnable classification token (CLS) is appended to the sequence of embedded patches. The input sequence to the transformer encoder is therefore defined as:

$$z_0 = [I_{class}; x_p^N E; \dots; x_p^1 E] + E_{pos} E \quad (2)$$

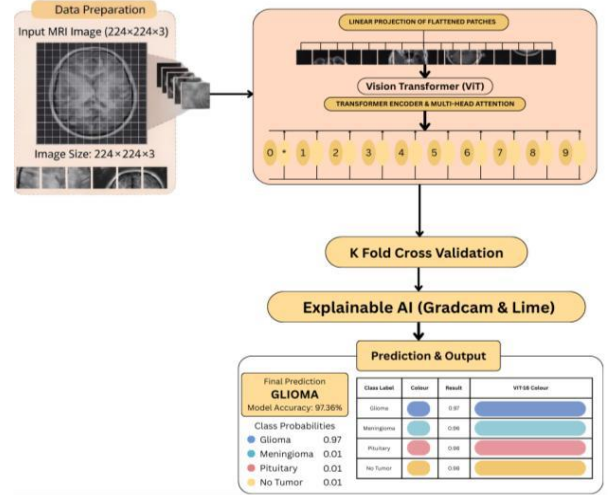


Fig. 3. Schematic of proposed ViT based brain tumor detection classifier.

where, x_p^N represents the patch embeddings, E denotes the embedding matrix, and E_{pos} corresponds to the positional encoding vectors [21].

The sequence is then passed through multiple Transformer encoder layers, each consisting of:

- Multi-Head Self-Attention (MSA)
- Feed-forward neural network (MLP)
- Layer normalization
- Residual connections

The output of the attention block is computed as:

$$Z'_l = MSA(LN(z_l - 1)) + z_{l-1} \quad l = 1 \dots L \quad (3)$$

and the output of the feed-forward network is given by:

$$z_l = MLP(LN(Z'_l \quad l = 1 \dots L)) \quad (4)$$

where, l denotes the index of the encoder layer [31].

The self-attention mechanism enables the model to learn relationships between different image patches. It computes attention scores using query (Q), key (K), and value (V) matrices as follows:

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right) V \quad (5)$$

where, d_k represents the dimensionality of the key vectors [22].

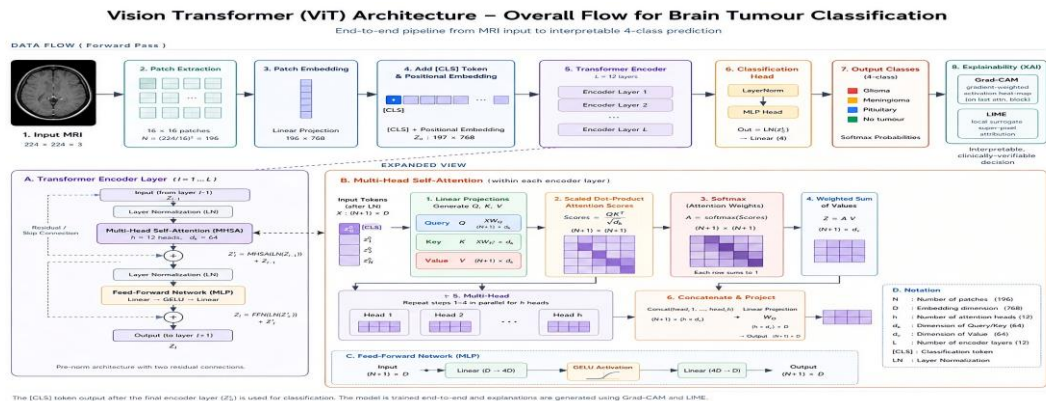


Fig. 4. Architecture of proposed ViT based brain tumor detection classifier.

Finally, the representation corresponding to the classification token is extracted from the encoder output and passed through a MLP head to generate the final classification output:

$$y = LN(z_l^0) \quad (6)$$

Due to its ability to model long-range dependencies and capture global contextual relationships among image regions, the Vision Transformer has shown strong performance in various computer vision tasks, including medical image analysis and brain tumor classification. Figure 4 shows the detailed architecture of the proposed Attention-Based Vision Transformer (VIT), highlighting its attention mechanisms, processing stages, and data flow to facilitate a clear understanding of the model and ensure reproducibility.

IV. RESULTS AND DISCUSSIONS

A. Model Implementation

The proposed VIT model has been implemented via TensorFlow Keras libraries under Python 3.7 using a Intel i7 11,800H, graphics card NVIDIA RTX 3060 6GB and RAM size of 64GB. The dataset consists of glioma, meningioma, pituitary, and no tumor classes. Here, each and every image are resized to 224×224 pixels and which is further normalized to ensure a consistent input distribution, this is where the data augmentation methods like rotations, and flips are used in improving the generalization of the model to diverse clinical samples. The VIT architecture is achieved after this preparation as the major diagnostic engine. Compared to traditional models, the VIT separates each scan of the MRI into a series of cross-sectional images of a fixed size and applies the self-attention to learn the overall spatial correlations enabling it to identify the intricate tumor texture and structural interactions within the entire image. In order to analyze and evaluate this architecture critically, a 5-fold cross validation strategy is implemented as the ultimate step of the training channel. To train the vision transformer network, Adam optimizer is applied with 120 epochs. A batch size of 64 and empirical learning rate of 0.001 are fixed to get fast computation and correct analysis.

B. Qualitative and Quantitative Performance

The performance of a classification model is measured by confusion matrix. The confusion matrix in Fig. 5 helps to achieve better analysis of the model through several architectures, followed by evaluation of performance metrics as Accuracy (Acc), Specificity (Spe), and Sensitivity (Sen), expressed as follows:

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN} \quad (7)$$

$$\text{Specificity} = \frac{TN}{TN+FP} \quad (8)$$

$$\text{Sensitivity(Recall)} = \frac{TP}{TP+FN} \quad (9)$$

where, true positive (TP), and true negative (TN) are represented by the diagonal elements, whereas the misclassification is highlighted by the off-diagonal elements, i.e., false positive (FP), and false negative (FN).

The confusion matrix for the implementation of VIT transformer highlights its good classification accuracy with most correct predictions along the diagonal of the confusion matrix & least miss classification. The model got almost all of the “No tumor” (537/544) and “Pituitary” (572/575) classes right, which shows that there are almost no false negatives in these important diagnoses. There are just a few minor confusions between “gliomas” and “Meningioma”. Therefore, the results prove the model’s high sensitivity and specificity, which indicates the VIT’s self-attention mechanism works to extract the different textural and spatial features needed for reliable multi-class brain tumor differentiation. Fig. 6 displays the testing accuracy and loss with respect to epochs for the proposed model under the training and testing phases. The results of the implementation show that the model has high diagnostic efficiency with a maximum training accuracy of 99.94% and a comprehensive testing accuracy of 98.51% throughout the 120-epoch training cycle.

C. Interpretable Results on MRI

In order to close the distance between clinical utility and high performance, the workflow incorporates the dual layer explainability using Gradient weighted Class Activation Mapping (Grad-CAM) and Local Interpretable Model-Agnostic Explanations (LIME). Grad-CAM produces activation heatmaps providing visualization on what the model focuses on the pathological regions in the brain, whereas the LIME will identify the individual super pixels and boundaries that contribute to the classification of the individual scans. The result of such interpretation tools has proven that the VIT model is capable of sufficiently focusing on tumor masses and not on unnecessary background noises, which creates a clear and trustworthy guide to the computer-assisted diagnosis. To make the 5-fold cross validation the last stage of the workflow, the study guarantees a stress test of the preprocessing on the entire set of data and a reliable diagnostic channel indicating a trustworthy model.

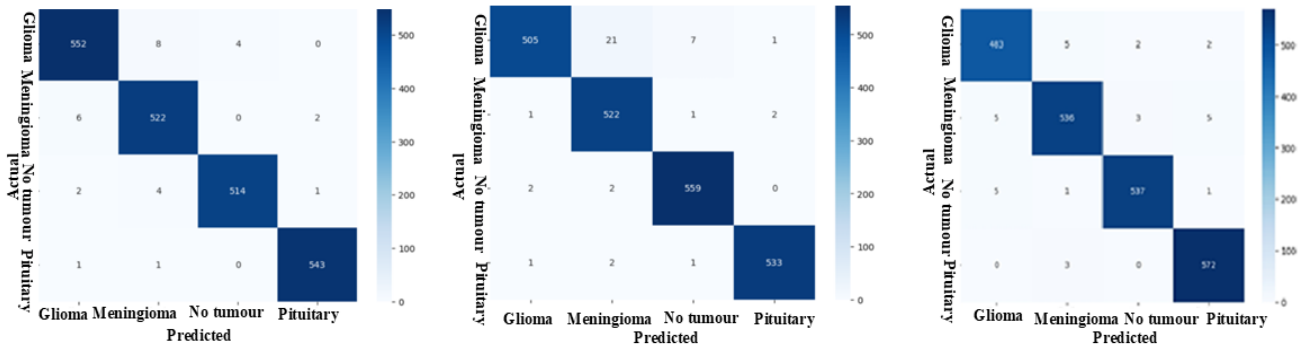


Fig. 5. Confusion matrix using (a) ResNet (b) DenseNet, (c) Proposed VIT Architecture.

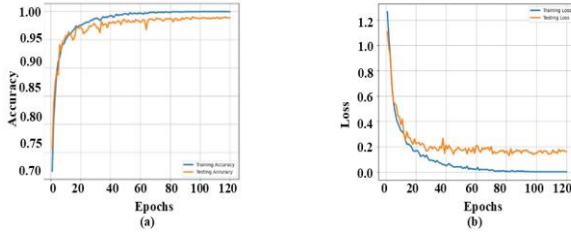


Fig. 6. Training- testing curves of the proposed model: (a) accuracy and (b) loss

Fig. 7 shows GRAD-CAM and LIME results on different MRI classes, that provides a very clear image on how the model is specifically interpreting each brain scan rather than merely guessing without addressing any context. The model believes that the red and orange parts are of more significance whereas the blue parts do not matter to make the choice. It is interesting how these areas of highlighting align almost accurately with the actual locations of tumors in the cases of Pituitary, Meningioma and gliomas. LIME demonstrates precisely which shapes or areas influenced prediction by separating the image into tiny pieces (super pixels) and drawing them in yellow. These tend to align with significant characteristics, such as odd tissue shapes or irregular edges which are often associated with tumors. The fact that the same spots are highlighted in both techniques indicates that the model is emphasizing true medical characteristics rather than noises and artifacts.

D. Comparative Results with Other Methods

Table I compare the classification results with various ML and DL models with the same dataset. Table II presents a

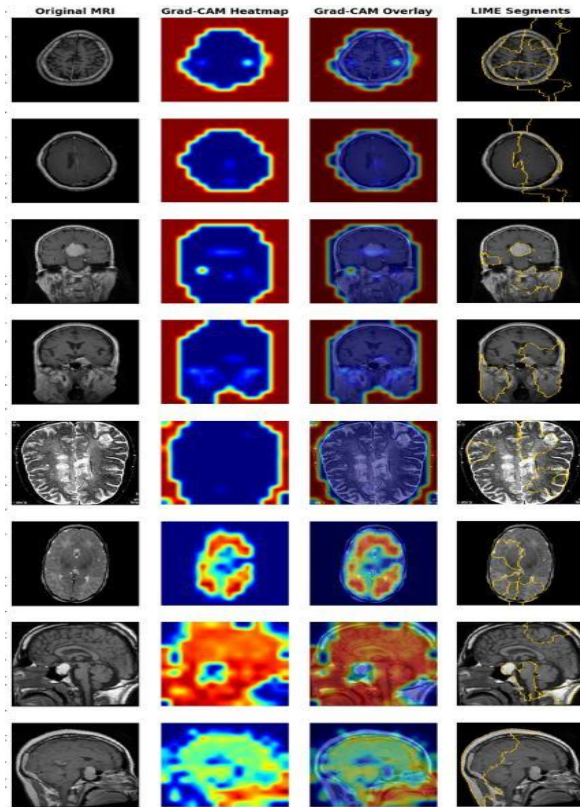


Fig. 7. GRAD-CAM and LIME results on different MRI classes.

comparative evaluation of various models on the same dataset. The comparative analysis shows that VIT architecture performs better than the other traditional CNNs on the all-evaluation matrices that are analyzed. Traditional CNN models like AlexNet and ResNet-50 had good accuracies like 92.18% and 94.25%, while DenseNet-201 performed much better than these and gave an accuracy of 95.84% but VIT showed the best performance with an accuracy of 98.51%. In addition to accuracy VIT proved its superior diagnostic reliability by achieving a specificity of 99.00% & sensitivity of 98.00% which proved that it is highly effective for correct identification of both healthy tissue and cancerous regions. Fig. 8. Shows the ablation Study of the Proposed VIT Pipeline components, showing the impact of preprocessing (Preproc.), data augmentation (Aug), cross-validation (CV), and explainability (XAI) on accuracy, sensitivity, and specificity. Table III Shows component-wise ablation analysis of the proposed attention-based VIT framework. The best classification accuracy is found to be $98.51 \pm 0.42\%$ with sensitivity of 98.00% and specificity of 99.00%, when the entire pipeline is used which included image preprocessing, data augmentation, 5-fold cross-validation and Explainable AI using Grad-CAM/LIME.

TABLE I
Comparison of various machine learning and deep learning models

Models	Acc	Spec	Sens
ResNet 50	94.25	94.00	93.00
AlexNet	92.18	92.00	91.00
DenseNet 201	95.84	96.00	95.00
Proposed	98.51	99.00	98.00

TABLE II
Comparison of Different Model Performances on the Same Dataset

Authors	Classifier	Acc	Spec	Sens
Alhassan <i>et al.</i> [17]	IV3TM	96.55	96.68	96.29
Ullah <i>et al.</i> [18]	DeepLabV3+	95.40	-	95.43
Hmidi <i>et al.</i> [19]	CNN	94.08	-	96.85
Sadad <i>et al.</i> [20]	ResNet50	93.10	-	-
Proposed Method	VIT	98.51	99.00	98.00

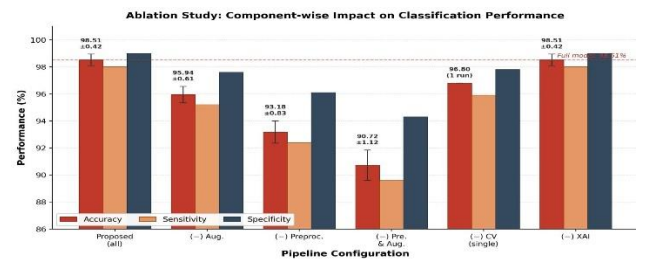


Fig. 8. Component-wise Impact of Pipeline Modules on Classification Performance

TABLE III

Configuration	Pre-proc.		Aug.	5-Fold CV	XAI	Acc (%)	Sens (%)	Spec (%)	Interp.
Proposed (all components)	✓		✓	✓	✓	98.51 ± 0.42	98.00	99.00	Yes
(-) Data Augmentation	✓		✗	✓	✓	95.94 ± 0.61	95.20	97.60	Yes
(-) Preprocessing (norm.)	✗		✓	✓	✓	93.18 ± 0.83	92.40	96.10	Yes
(-) Preprocessing & Aug.	✗		✗	✓	✓	90.72 ± 1.12	89.60	94.30	Yes
(-) 5-Fold CV (single split)	✓		✓	✗	✓	96.80 (1 run)	95.90	97.80	Yes
(-) Explainability (CAM/LIME)	✓		✓	✓	✗	98.51 ± 0.42	98.00	99.00	No

V. CONCLUSION

This paper introduces a method that classifies brain tumors employing VIT, which outperforms classical deep learnings model by its capability to identify the overall features of contextual links of MRI images. The proposed methodology with built-in comprehensive preprocessing, data augmentation and k-fold cross validation provides stable and unbiased cross evaluation of all the data. The model had good diagnostic validity, as well as excellent sensitivity and specificity, which means that it is effective in differentiating various forms of tumors and non-cancerous cases. The major drawback of traditional deep learning systems: the black-box nature has been successfully resolved by introducing visual and interpretable features of XAI (Grad-CAM and LIME) methods that can be used to understand the decision of a model, making the system more credible and transparent to clinicians. In sum, the suggested system is a scalable, precise and interpretable approach to automated brain tumor detection, and has potential major clinical benefits in real-life clinical use and future developments in intelligent medical imaging systems.

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